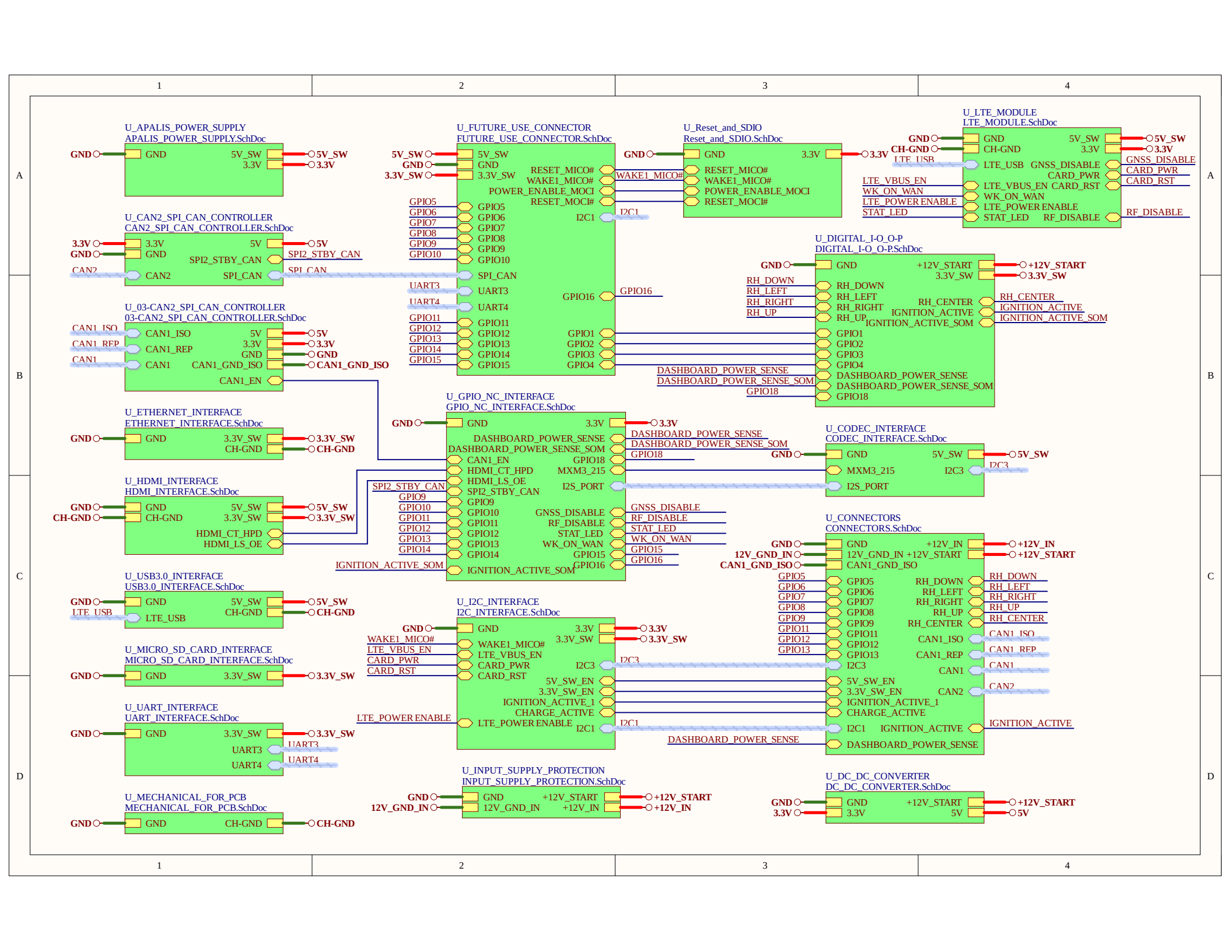
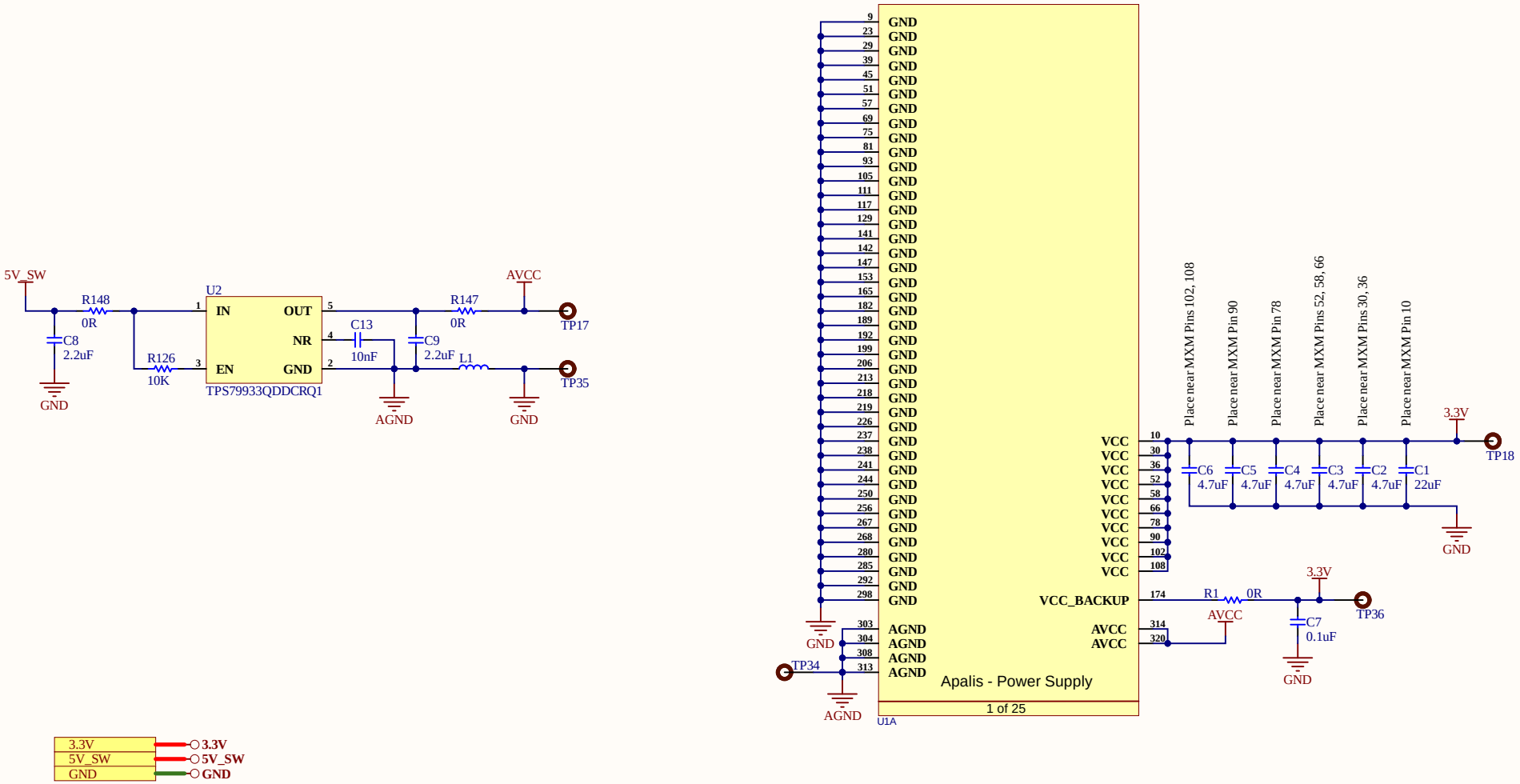
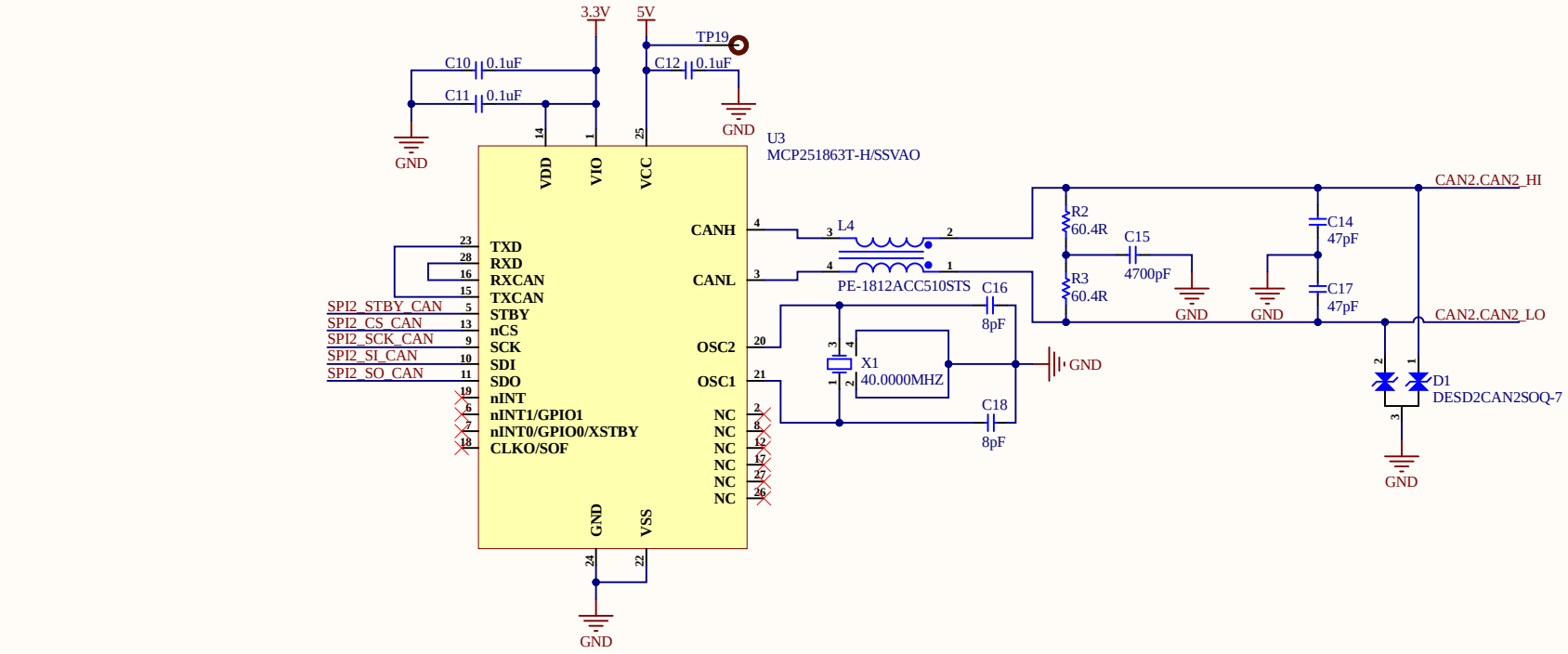
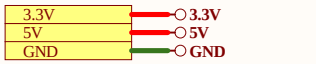
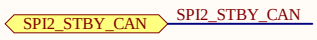

U_APALIS_POWER_SUPPLY
APALIS_POWER_SUPPLY.SchDoc
GND
5V_SW
3.3V
U_FUTURE_USE_CONNECTOR
FUTURE_USE_CONNECTOR.SchDoc
RESET_MICO#
WAKE1_MICO#
POWER_ENABLE_MOCI
RESET_MOCI#
I2C1
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
SPI_CAN
UART3
UART4
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO1
GPIO2
GPIO3
GPIO4
U_Reset_and_SDIO
Reset_and_SDIO.SchDoc
U_LTE_MODULE
LTE_MODULE.SchDoc
LTE_USB
GNSS_DISABLE
CARD_PWR
CARD_RST
LTE_VBUS_EN
WK_ON_WAN
LTE_POWER ENABLE
STAT_LED
RF_DISABLE
U_CAN2_SPI_CAN_CONTROLLER
CAN2_SPI_CAN_CONTROLLER.SchDoc
SPI2_STBY_CAN
CAN2
U_03-CAN2_SPI_CAN_CONTROLLER
03-CAN2_SPI_CAN_CONTROLLER.SchDoc
CAN1_ISO
CAN1_REP
CAN1
CAN1_GND_ISO
CAN1_EN
U_DIGITAL_I-O_O-P
DIGITAL_I-O_O-P.SchDoc
+12V_START
3.3V_SW
RH_DOWN
RH_LEFT
RH_RIGHT
RH_UP
RH_CENTER
IGNITION_ACTIVE
IGNITION_ACTIVE_SOM
DASHBOARD_POWER_SENSE
DASHBOARD_POWER_SENSE_SOM
U_ETHERNET_INTERFACE
ETHERNET_INTERFACE.SchDoc
CH-GND
U_GPIO_NC_INTERFACE
GPIO_NC_INTERFACE.SchDoc
DASHBOARD_POWER_SENSE
DASHBOARD_POWER_SENSE_SOM
GPIO18
HDMI_CT_HPD
HDMI_LS_OE
MXM3_215
I2S_PORT
SPI2_STBY_CAN
GNSS_DISABLE
RF_DISABLE
STAT_LED
WK_ON_WAN
IGNITION_ACTIVE_SOM
U_CODEC_INTERFACE
CODEC_INTERFACE.SchDoc
I2C3
U_HDMI_INTERFACE
HDMI_INTERFACE.SchDoc
U_CONNECTORS
CONNECTORS.SchDoc
12V_GND_IN
CAN1_GND_ISO
U_USB3.0_INTERFACE
USB3.0_INTERFACE.SchDoc
U_I2C_INTERFACE
I2C_INTERFACE.SchDoc
5V_SW_EN
3.3V_SW_EN
IGNITION_ACTIVE_1
CHARGE_ACTIVE
U_MICRO_SD_CARD_INTERFACE
MICRO_SD_CARD_INTERFACE.SchDoc
U_UART_INTERFACE
UART_INTERFACE.SchDoc
U_INPUT_SUPPLY_PROTECTION
INPUT_SUPPLY_PROTECTION.SchDoc
U_DC_DC_CONVERTER
DC_DC_CONVERTER.SchDoc
U_MECHANICAL_FOR_PCB
MECHANICAL_FOR_PCB.SchDoc

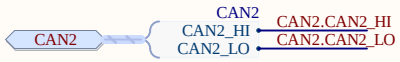
TARFORM

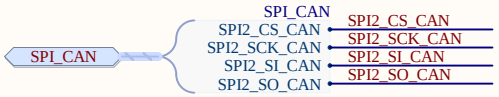

5V_SW
R148
0R
C8
2.2uF
GND
U2
IN
OUT
NR
EN
GND
TPS79933QDDCRQ1
R126
10K
C13
10nF
C9
2.2uF
L1
AGND
R147
0R
AVCC
TP17
TP35
GND
GND
GND

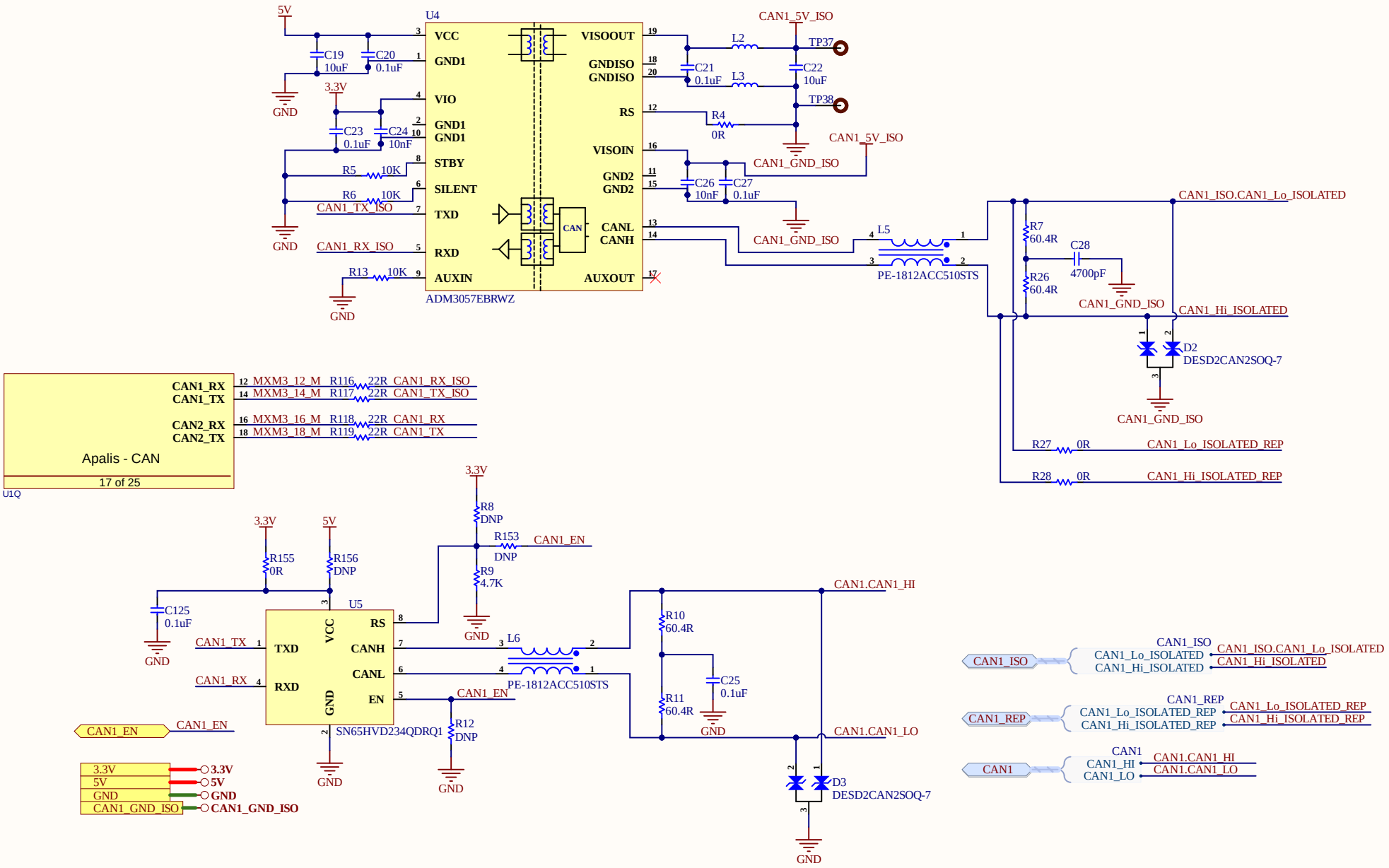

3.3V
5V
TP19
C10 0.1uF
C11 0.1uF
C12 0.1uF
GND
U3
MCP251863T-H/SSVAO
VDD
VIO
VCC
GND
VSS
TXD
RXD
RXCAN
TXCAN
STBY
nCS
SCK
SDI
SDO
nINT
nINT1/GPIO1
nINT0/GPIO0/XSTBY
CLKO/SOF
CANH
CANL
OSC2
OSC1
NC
SPI2_STBY_CAN
SPI2_CS_CAN
SPI2_SCK_CAN
SPI2_SI_CAN
SPI2_SO_CAN
L4
PE-1812ACC510STS
R2 60.4R
R3 60.4R
C15 4700pF
C16 8pF
C18 8pF
X1 40.0000MHZ
C14 47pF
C17 47pF
CAN2.CAN2_HI
CAN2.CAN2_LO
D1
DESD2CAN2SOQ-7


3.3V
5V
GND
3.3V
5V
GND

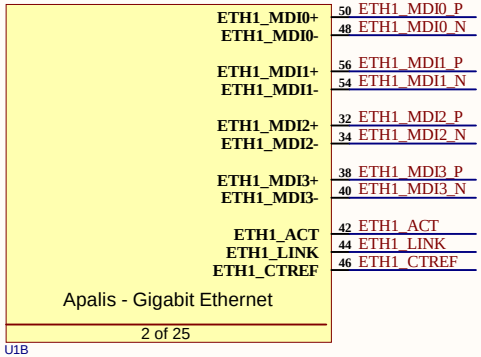

SPI2_STBY_CAN
SPI2_STBY_CAN

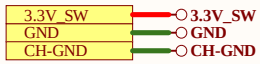

CAN2
CAN2
CAN2_HI
CAN2_LO
CAN2.CAN2_HI
CAN2.CAN2_LO

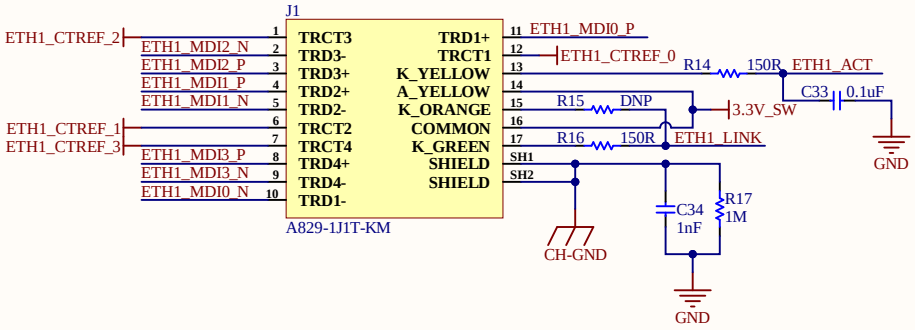

SPI_CAN
SPI_CAN
SPI2_CS_CAN
SPI2_SCK_CAN
SPI2_SI_CAN
SPI2_SO_CAN
SPI2_CS_CAN
SPI2_SCK_CAN
SPI2_SI_CAN
SPI2_SO_CAN

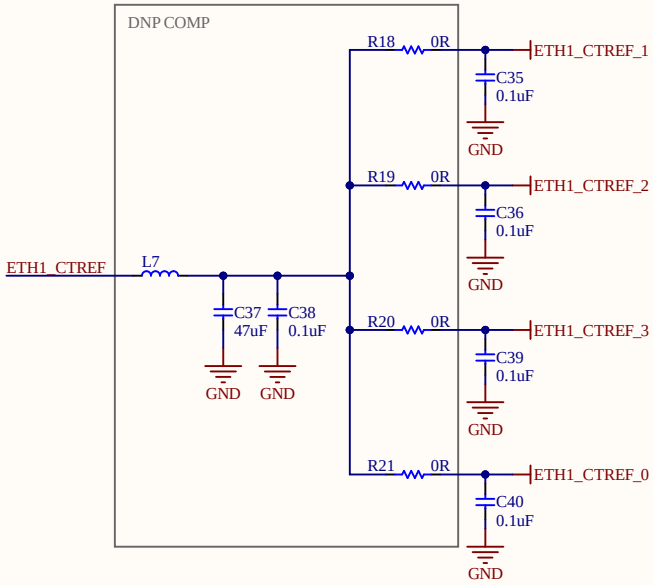

U4
ADM3057EBRWZ
VCC
GND1
VIO
GND1
GND1
STBY
SILENT
TXD
RXD
AUXIN
VISOOUT
GNDISO
GNDISO
RS
VISOIN
GND2
GND2
CANL
CANH
AUXOUT
CAN
5V
3.3V
GND
C19 10uF
C20 0.1uF
C23 0.1uF
C24 10nF
R5 10K
R6 10K
R13 10K
CAN1_TX_ISO
CAN1_RX_ISO
C21 0.1uF
C22 10uF
C26 10nF
C27 0.1uF
L2
L3
R4 0R
TP37
TP38
CAN1_5V_ISO
CAN1_GND_ISO
L5
PE-1812ACC510STS
R7 60.4R
R26 60.4R
C28 4700pF
CAN1_ISO.CAN1_Lo_ISOLATED
CAN1_Hi_ISOLATED
D2
DESD2CAN2SOQ-7
R27 0R
R28 0R
CAN1_Lo_ISOLATED_REP
CAN1_Hi_ISOLATED_REP
CAN1_RX
CAN1_TX
CAN2_RX
CAN2_TX
MXM3_12_M
MXM3_14_M
MXM3_16_M
MXM3_18_M
R116 22R CAN1_RX_ISO
R117 22R CAN1_TX_ISO
R118 22R CAN1_RX
R119 22R CAN1_TX
Apalis - CAN
17 of 25
U1Q
U5
SN65HVD234QDRQ1
R155 0R
R156 DNP
R8 DNP
R153 DNP
R9 4.7K
R12 DNP
C125 0.1uF
L6
PE-1812ACC510STS
R10 60.4R
R11 60.4R
C25 0.1uF
CAN1_EN
CAN1.CAN1_HI
CAN1.CAN1_LO
D3
DESD2CAN2SOQ-7
CAN1_ISO
CAN1_REP
CAN1
3.3V
5V
GND
CAN1_GND_ISO

ETHERNET INTERFACE

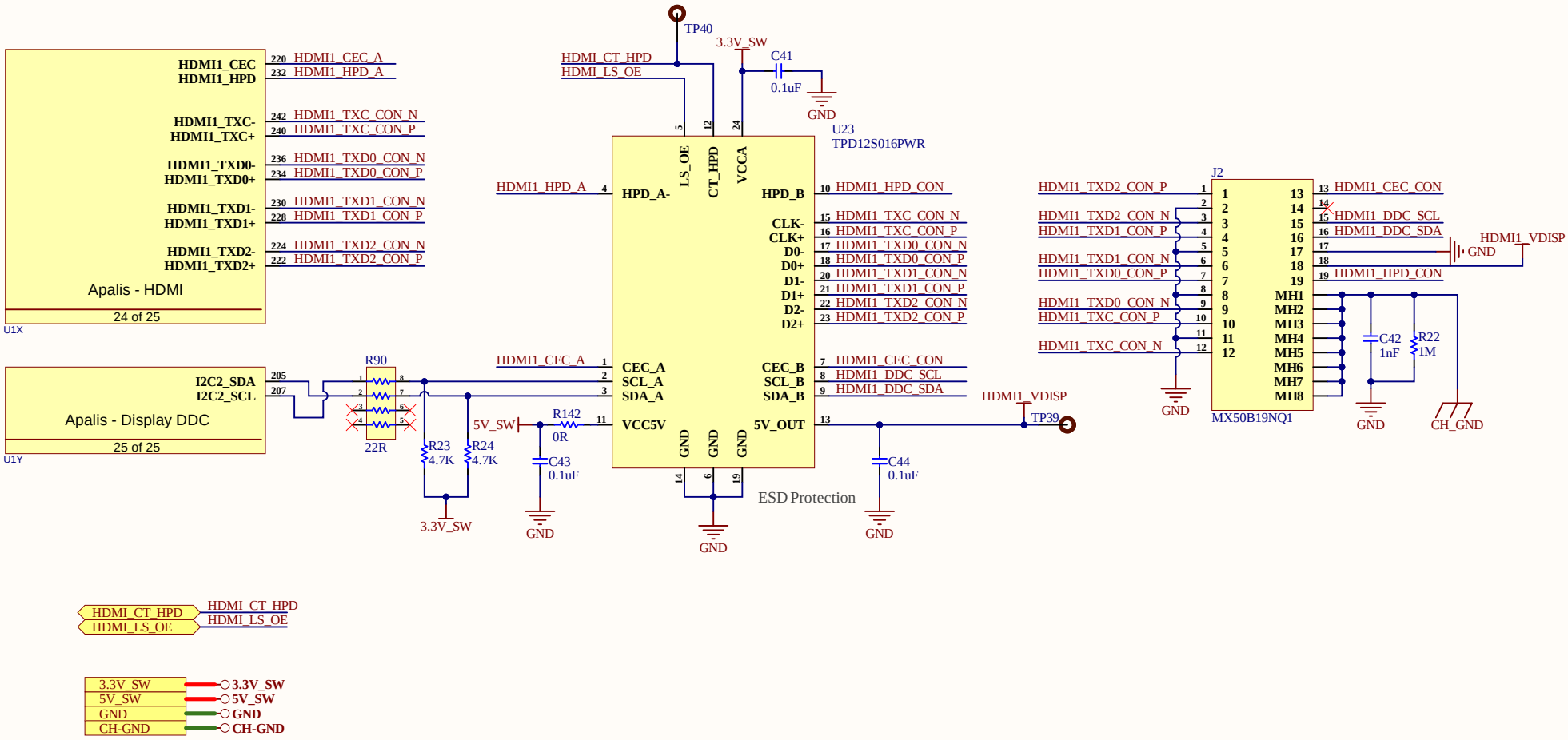




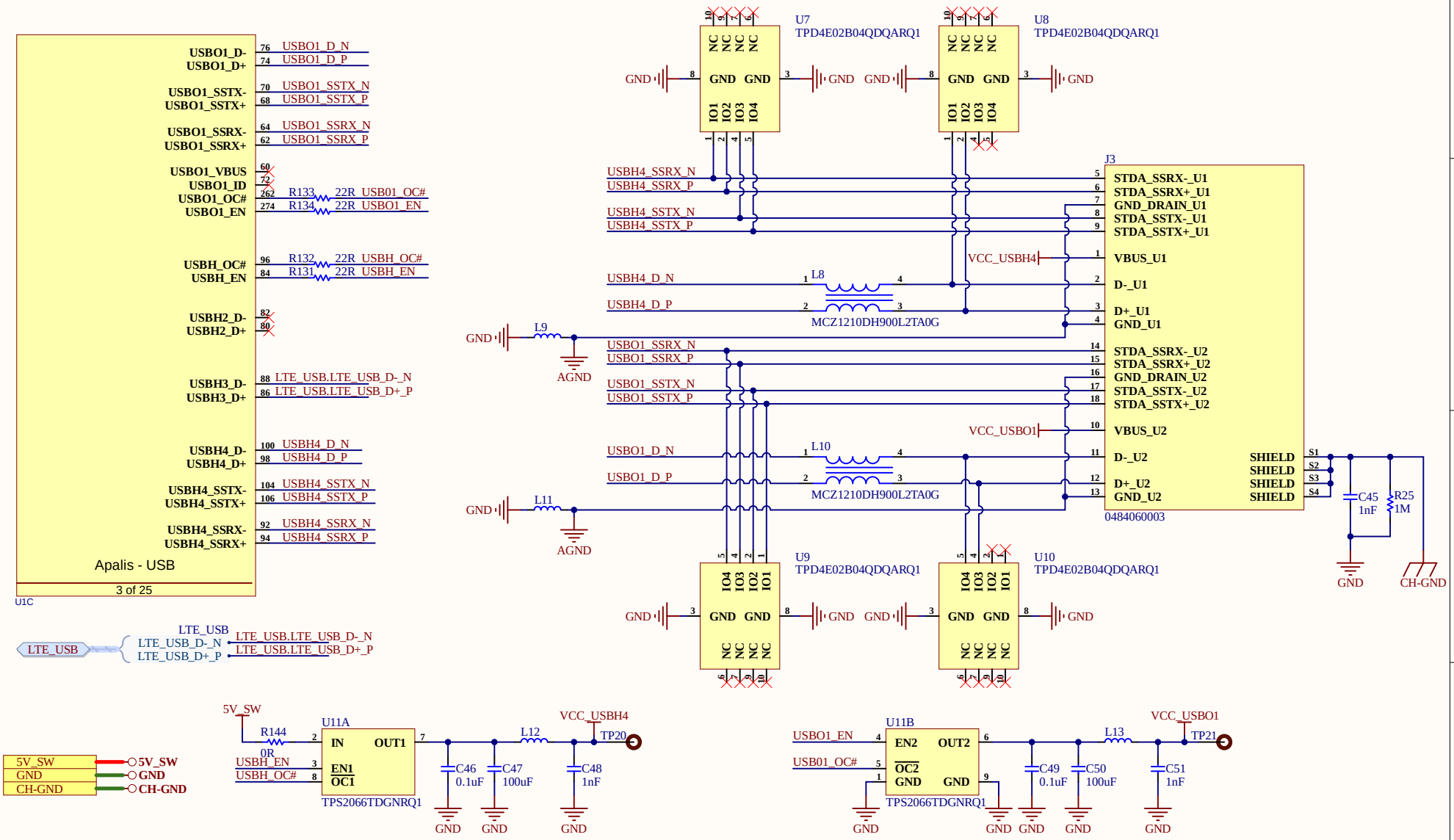




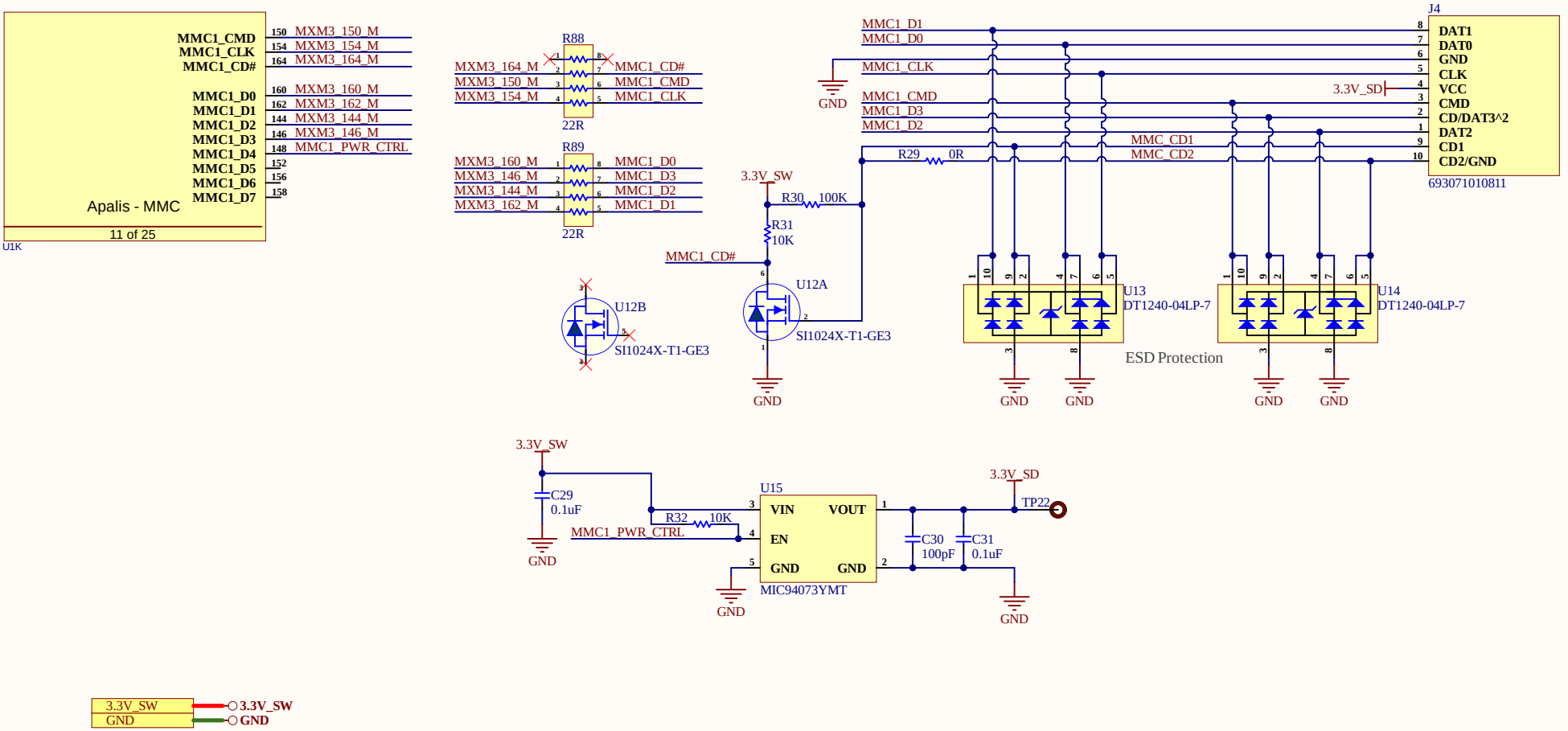
HDMI INTERFACE



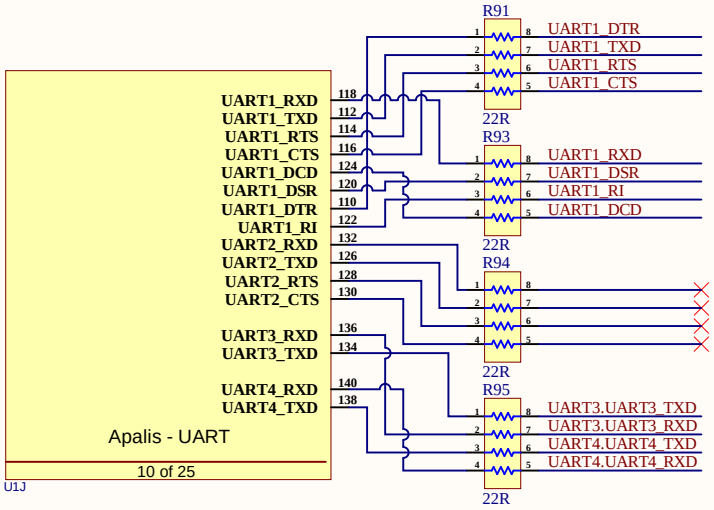
USB3.0 INTERFACE

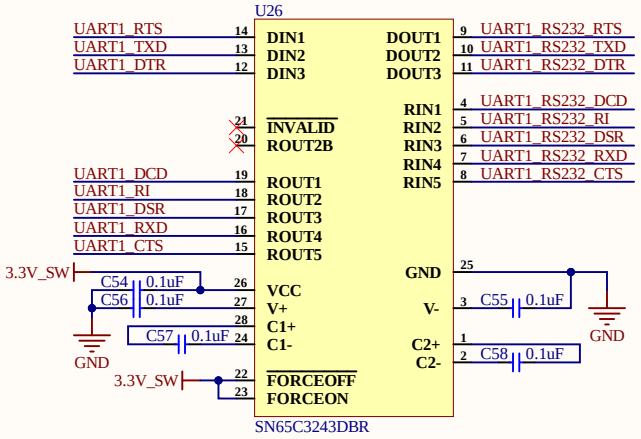


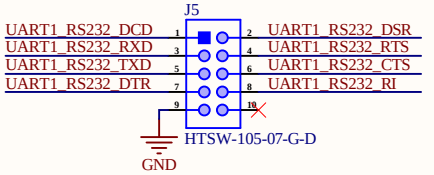
MICRO SD CARD INTERFACE



UART INTERFACE







CONNECTOR PENDING

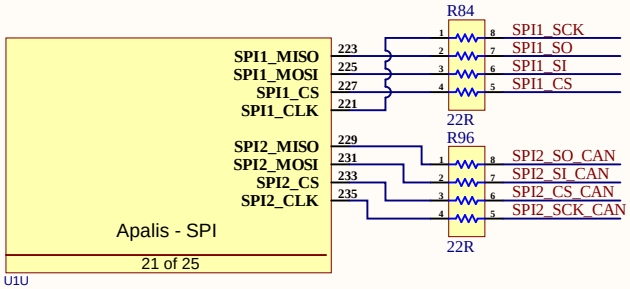
UART1 - FOR DEBUG
UART2 - FOR BT
UART3 - FOR FUTURE USE
UART4 - FOR FUTURE USE

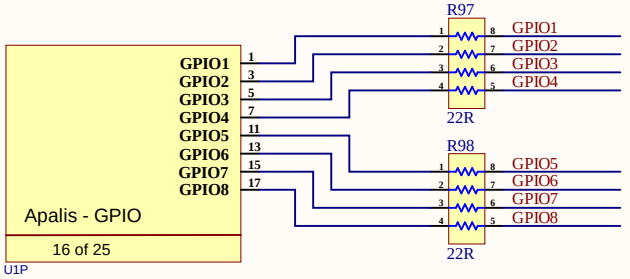


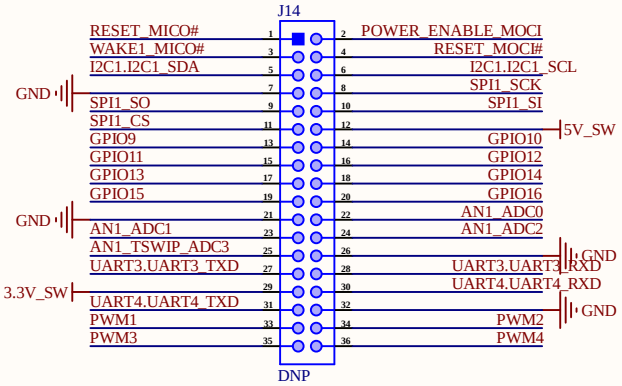


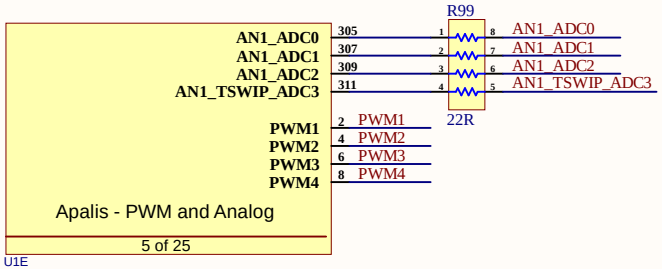


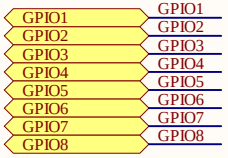
FUTURE USE CONNECTOR

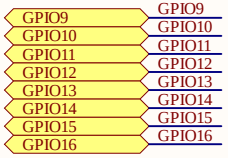




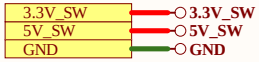


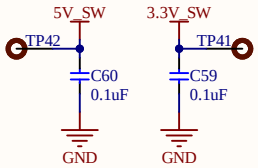




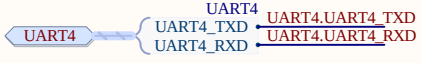




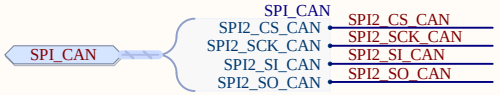


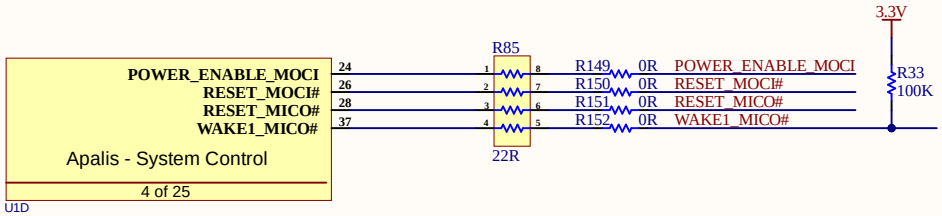


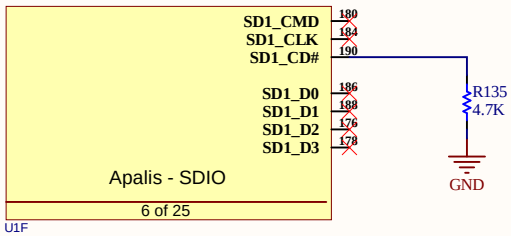


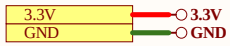


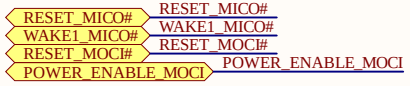


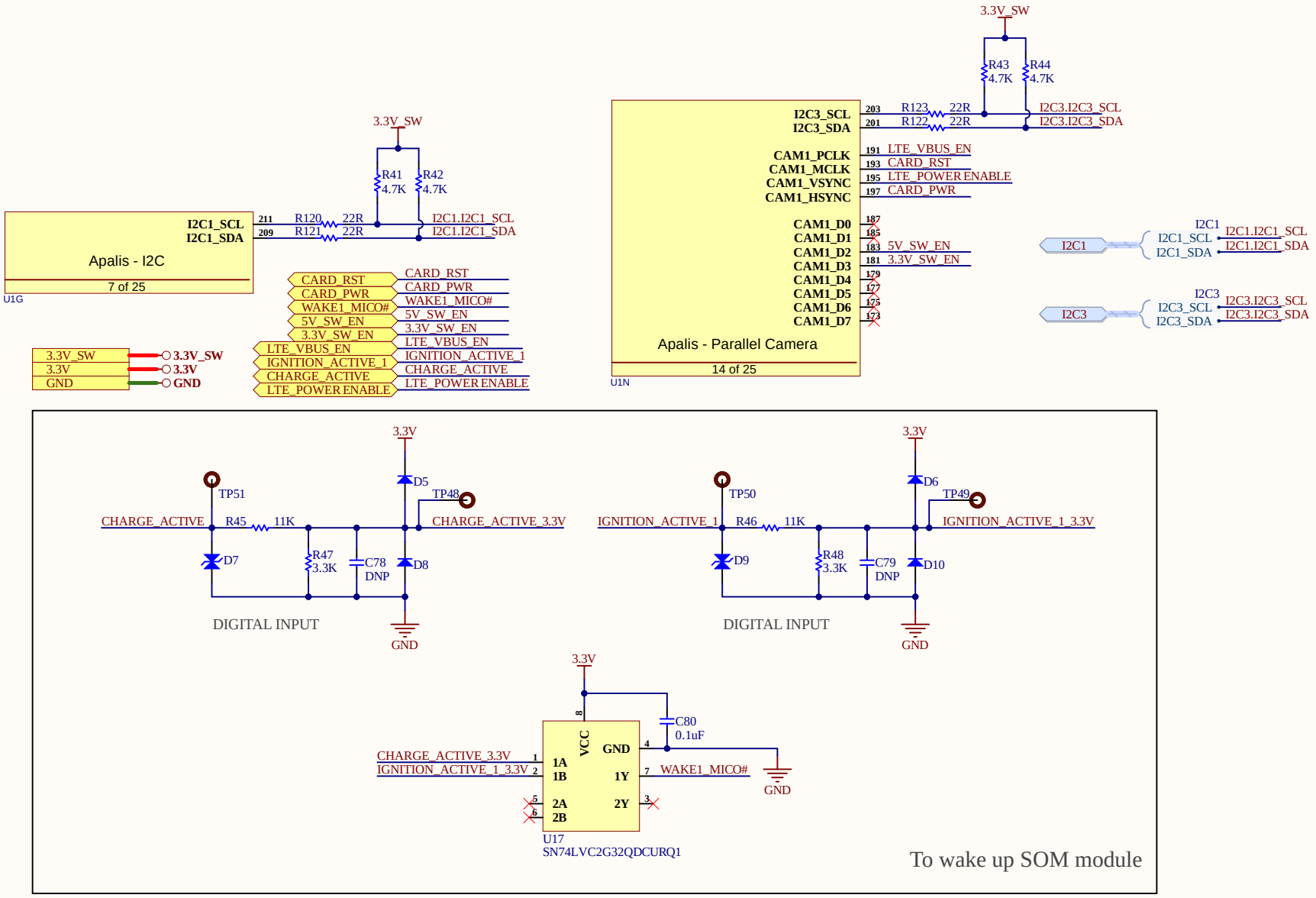


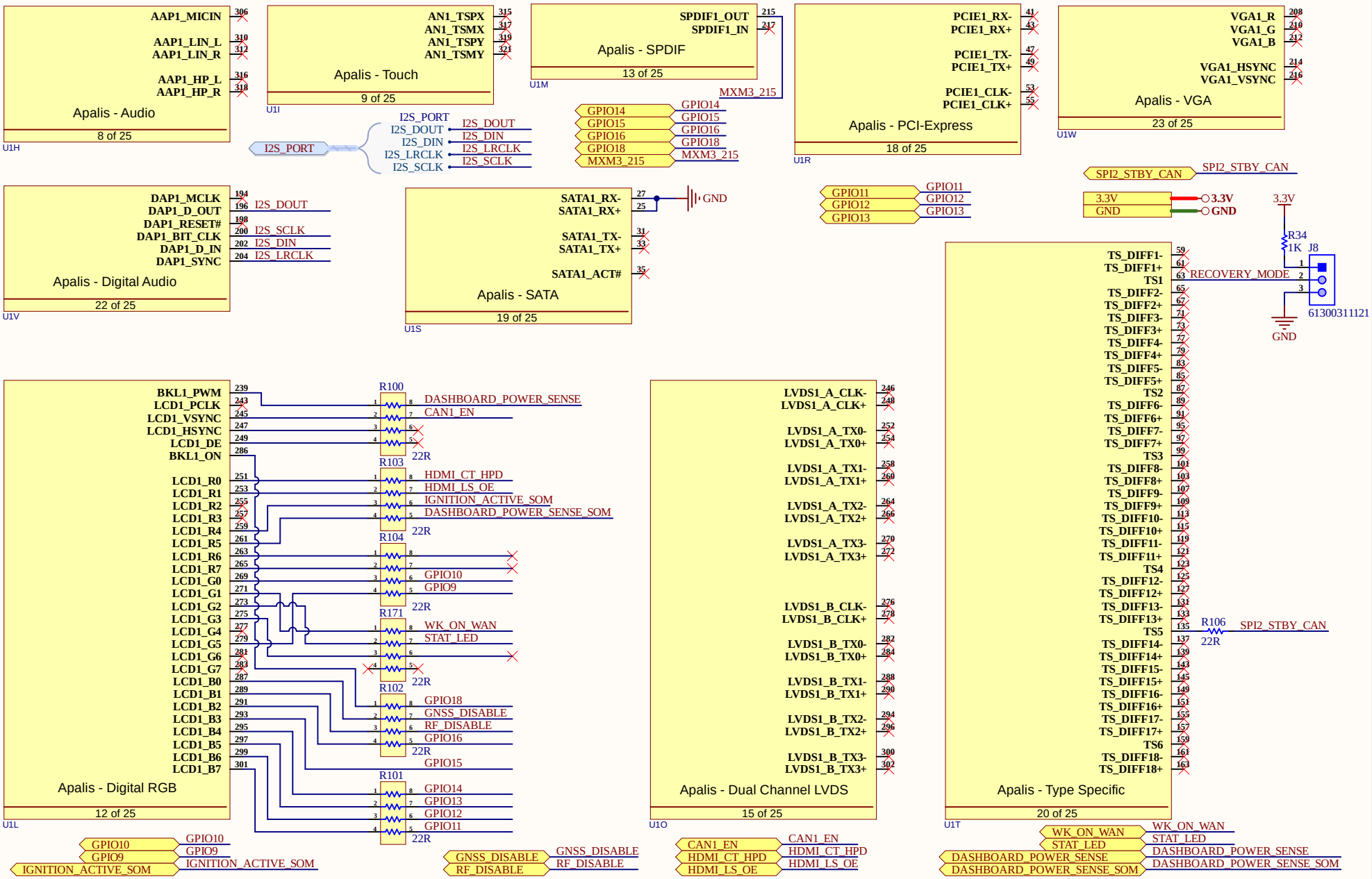

POWER_ENABLE_MOCI
RESET_MOCI#
RESET_MICO#
WAKE1_MICO#
Apalis - System Control
4 of 25
U1D
24
26
28
37
R85
22R
R149 0R POWER_ENABLE_MOCI
R150 0R RESET_MOCI#
R151 0R RESET_MICO#
R152 0R WAKE1_MICO#
3.3V
R33
100K

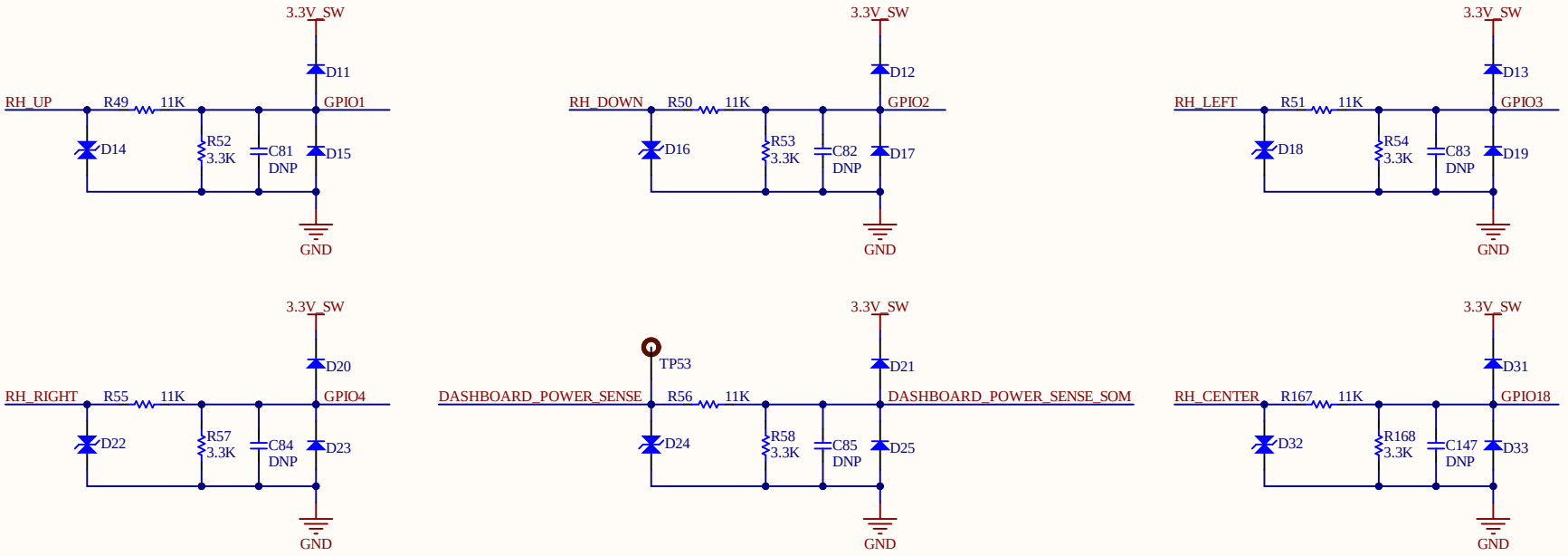

SD1_CMD
SD1_CLK
SD1_CD#
SD1_D0
SD1_D1
SD1_D2
SD1_D3
Apalis - SDIO
6 of 25
U1F
180
184
190
186
188
176
178
R135
4.7K
GND


3.3V
GND
3.3V
GND

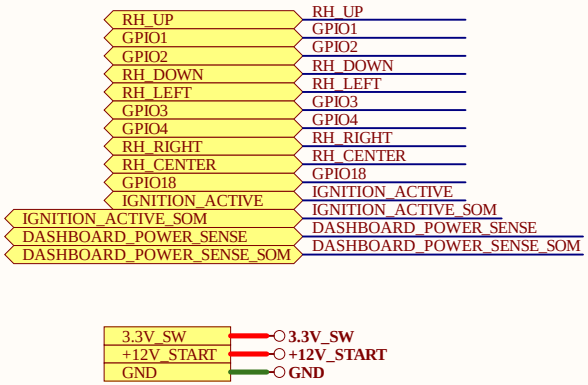

RESET_MICO#
WAKE1_MICO#
RESET_MOCI#
POWER_ENABLE_MOCI
RESET_MICO#
WAKE1_MICO#
RESET_MOCI#
POWER_ENABLE_MOCI

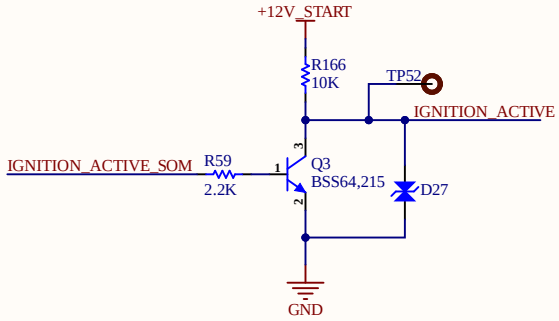

3.3V_SW
R43
4.7K
R44
4.7K
I2C3_SCL
I2C3_SDA
203
201
R123
22R
R122
22R
I2C3.I2C3_SCL
I2C3.I2C3_SDA
CAM1_PCLK
CAM1_MCLK
CAM1_VSYNC
CAM1_HSYNC
191
193
195
197
LTE_VBUS_EN
CARD_RST
LTE_POWER ENABLE
CARD_PWR
CAM1_D0
CAM1_D1
CAM1_D2
CAM1_D3
CAM1_D4
CAM1_D5
CAM1_D6
CAM1_D7
187
185
183
181
179
177
175
173
5V_SW_EN
3.3V_SW_EN
Apalis - Parallel Camera
14 of 25
U1N
I2C1
I2C1_SCL
I2C1_SDA
I2C1.I2C1_SCL
I2C1.I2C1_SDA
I2C3
I2C3_SCL
I2C3_SDA
I2C3.I2C3_SCL
I2C3.I2C3_SDA
3.3V_SW
R41
4.7K
R42
4.7K
I2C1_SCL
I2C1_SDA
211
209
R120
22R
R121
22R
I2C1.I2C1_SCL
I2C1.I2C1_SDA
Apalis - I2C
7 of 25
U1G
CARD_RST
CARD_PWR
WAKE1_MICO#
5V_SW_EN
3.3V_SW_EN
LTE_VBUS_EN
IGNITION_ACTIVE_1
CHARGE_ACTIVE
LTE_POWER ENABLE
3.3V_SW
3.3V
GND
3.3V_SW
3.3V
GND
3.3V
TP51
D5
TP48
CHARGE_ACTIVE
R45
11K
CHARGE_ACTIVE_3.3V
D7
R47
3.3K
C78
DNP
D8
GND
DIGITAL INPUT
3.3V
TP50
D6
TP49
IGNITION_ACTIVE_1
R46
11K
IGNITION_ACTIVE_1_3.3V
D9
R48
3.3K
C79
DNP
D10
GND
DIGITAL INPUT
3.3V
C80
0.1uF
CHARGE_ACTIVE_3.3V
IGNITION_ACTIVE_1_3.3V
VCC
GND
1A
1B
1Y
2A
2B
2Y
WAKE1_MICO#
GND
U17
SN74LVC2G32QDCURQ1
To wake up SOM module


AAP1_MICIN
AAP1_LIN_L
AAP1_LIN_R
AAP1_HP_L
AAP1_HP_R
Apalis - Audio
8 of 25
U1H
AN1_TSPX
AN1_TSMX
AN1_TSPY
AN1_TSMY
Apalis - Touch
9 of 25
U1I
SPDIF1_OUT
SPDIF1_IN
Apalis - SPDIF
13 of 25
U1M
MXM3_215
PCIE1_RX-
PCIE1_RX+
PCIE1_TX-
PCIE1_TX+
PCIE1_CLK-
PCIE1_CLK+
Apalis - PCI-Express
18 of 25
U1R
VGA1_R
VGA1_G
VGA1_B
VGA1_HSYNC
VGA1_VSYNC
Apalis - VGA
23 of 25
U1W
I2S_PORT
I2S_DOUT
I2S_DIN
I2S_LRCLK
I2S_SCLK
GPIO14
GPIO15
GPIO16
GPIO18
MXM3_215
SPI2_STBY_CAN
3.3V
GND
DAP1_MCLK
DAP1_D_OUT
DAP1_RESET#
DAP1_BIT_CLK
DAP1_D_IN
DAP1_SYNC
Apalis - Digital Audio
22 of 25
U1V
SATA1_RX-
SATA1_RX+
SATA1_TX-
SATA1_TX+
SATA1_ACT#
Apalis - SATA
19 of 25
U1S
GPIO11
GPIO12
GPIO13
R34
1K
J8
61300311121
RECOVERY_MODE
TS_DIFF1-
TS_DIFF1+
TS1
TS_DIFF2-
TS_DIFF2+
TS_DIFF3-
TS_DIFF3+
TS_DIFF4-
TS_DIFF4+
TS_DIFF5-
TS_DIFF5+
TS2
TS_DIFF6-
TS_DIFF6+
TS_DIFF7-
TS_DIFF7+
TS3
TS_DIFF8-
TS_DIFF8+
TS_DIFF9-
TS_DIFF9+
TS_DIFF10-
TS_DIFF10+
TS_DIFF11-
TS_DIFF11+
TS4
TS_DIFF12-
TS_DIFF12+
TS_DIFF13-
TS_DIFF13+
TS5
TS_DIFF14-
TS_DIFF14+
TS_DIFF15-
TS_DIFF15+
TS_DIFF16-
TS_DIFF16+
TS_DIFF17-
TS_DIFF17+
TS6
TS_DIFF18-
TS_DIFF18+
R106
22R
Apalis - Type Specific
20 of 25
U1T
BKL1_PWM
LCD1_PCLK
LCD1_VSYNC
LCD1_HSYNC
LCD1_DE
BKL1_ON
LCD1_R0
LCD1_R1
LCD1_R2
LCD1_R3
LCD1_R4
LCD1_R5
LCD1_R6
LCD1_R7
LCD1_G0
LCD1_G1
LCD1_G2
LCD1_G3
LCD1_G4
LCD1_G5
LCD1_G6
LCD1_G7
LCD1_B0
LCD1_B1
LCD1_B2
LCD1_B3
LCD1_B4
LCD1_B5
LCD1_B6
LCD1_B7
Apalis - Digital RGB
12 of 25
U1L
R100
R103
R104
R171
R102
R101
22R
DASHBOARD_POWER_SENSE
CAN1_EN
HDMI_CT_HPD
HDMI_LS_OE
IGNITION_ACTIVE_SOM
DASHBOARD_POWER_SENSE_SOM
GPIO10
GPIO9
WK_ON_WAN
STAT_LED
GPIO18
GNSS_DISABLE
RF_DISABLE
GPIO16
GPIO15
GPIO14
GPIO13
GPIO12
GPIO11
LVDS1_A_CLK-
LVDS1_A_CLK+
LVDS1_A_TX0-
LVDS1_A_TX0+
LVDS1_A_TX1-
LVDS1_A_TX1+
LVDS1_A_TX2-
LVDS1_A_TX2+
LVDS1_A_TX3-
LVDS1_A_TX3+
LVDS1_B_CLK-
LVDS1_B_CLK+
LVDS1_B_TX0-
LVDS1_B_TX0+
LVDS1_B_TX1-
LVDS1_B_TX1+
LVDS1_B_TX2-
LVDS1_B_TX2+
LVDS1_B_TX3-
LVDS1_B_TX3+
Apalis - Dual Channel LVDS
15 of 25
U1O

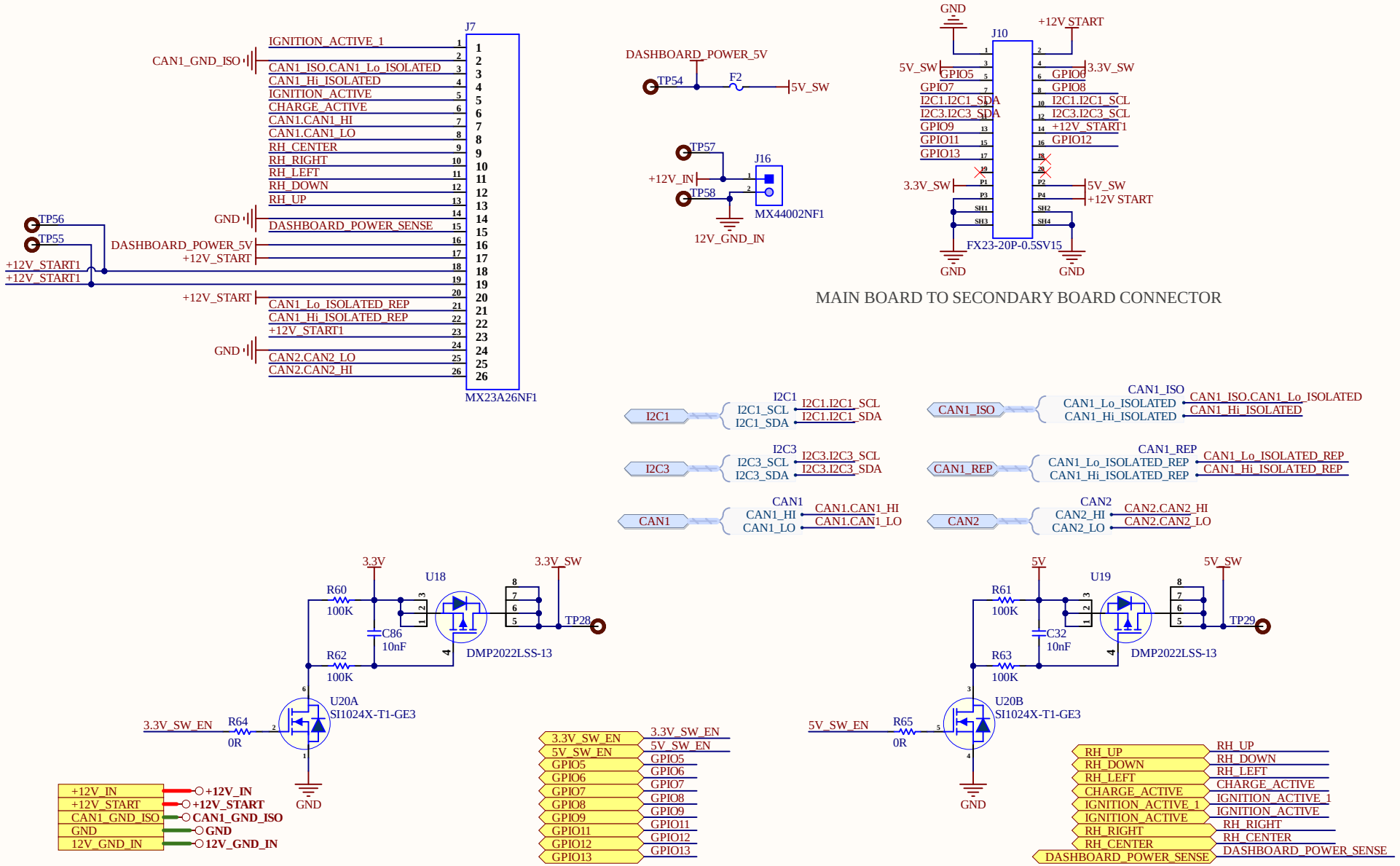


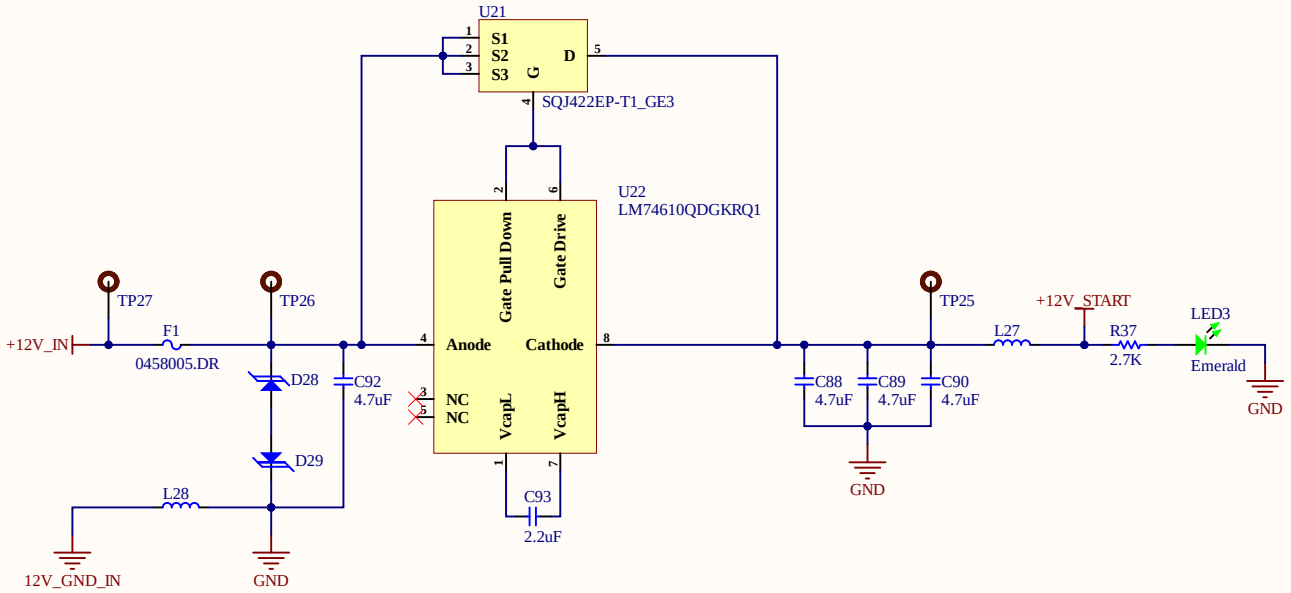
DIGITAL INPUT

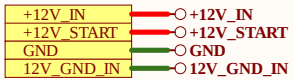


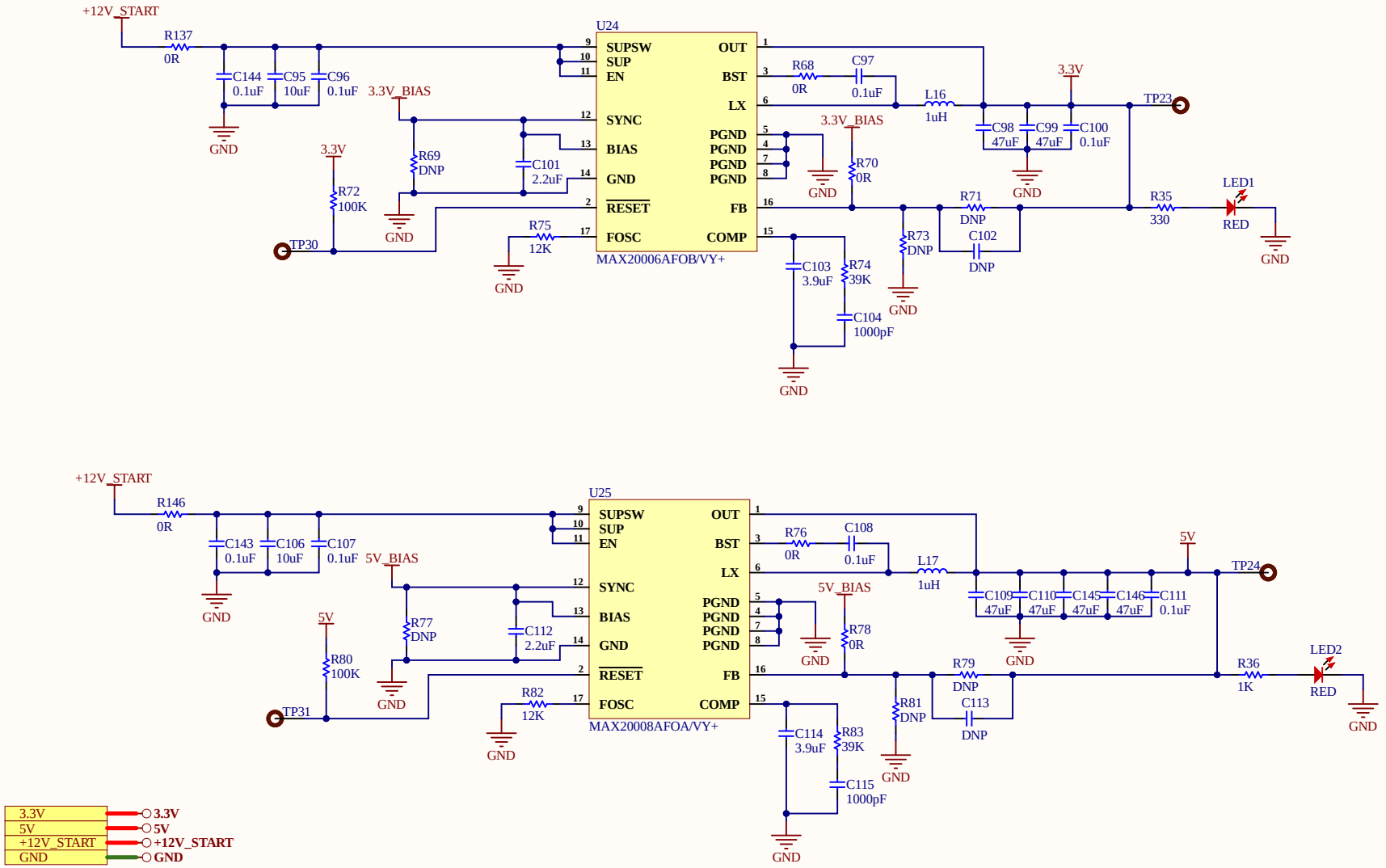


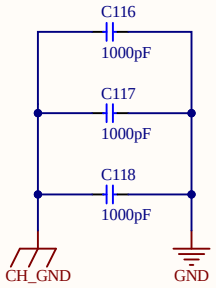
DIGITAL OUTPUT


J7
IGNITION_ACTIVE_1
CAN1_GND_ISO
CAN1_ISO.CAN1_Lo_ISOLATED
CAN1_Hi_ISOLATED
IGNITION_ACTIVE
CHARGE_ACTIVE
CAN1.CAN1_HI
CAN1.CAN1_LO
RH_CENTER
RH_RIGHT
RH_LEFT
RH_DOWN
RH_UP
GND
DASHBOARD_POWER_SENSE
TP56
TP55
DASHBOARD_POWER_5V
+12V_START
+12V_START1
+12V_START1
+12V_START
CAN1_Lo_ISOLATED_REP
CAN1_Hi_ISOLATED_REP
+12V_START1
GND
CAN2.CAN2_LO
CAN2.CAN2_HI
MX23A26NF1
DASHBOARD_POWER_5V
TP54
F2
5V_SW
TP57
J16
+12V_IN
TP58
MX44002NF1
12V_GND_IN
GND
+12V_START
J10
5V_SW
3.3V_SW
GPIO5
GPIO6
GPIO7
GPIO8
I2C1.I2C1_SDA
I2C1.I2C1_SCL
I2C3.I2C3_SDA
I2C3.I2C3_SCL
GPIO9
+12V_START1
GPIO11
GPIO12
GPIO13
3.3V_SW
5V_SW
+12V START
FX23-20P-0.5SV15
GND
GND
MAIN BOARD TO SECONDARY BOARD CONNECTOR
I2C1
I2C1_SCL
I2C1_SDA
I2C1.I2C1_SCL
I2C1.I2C1_SDA
CAN1_ISO
CAN1_Lo_ISOLATED
CAN1_Hi_ISOLATED
CAN1_ISO.CAN1_Lo_ISOLATED
CAN1_Hi_ISOLATED
I2C3
I2C3_SCL
I2C3_SDA
I2C3.I2C3_SCL
I2C3.I2C3_SDA
CAN1_REP
CAN1_Lo_ISOLATED_REP
CAN1_Hi_ISOLATED_REP
CAN1_Lo_ISOLATED_REP
CAN1_Hi_ISOLATED_REP
CAN1
CAN1_HI
CAN1_LO
CAN1.CAN1_HI
CAN1.CAN1_LO
CAN2
CAN2_HI
CAN2_LO
CAN2.CAN2_HI
CAN2.CAN2_LO
3.3V
R60
100K
U18
3.3V_SW
TP28
C86
10nF
R62
100K
DMP2022LSS-13
U20A
SI1024X-T1-GE3
3.3V_SW_EN
R64
0R
GND
5V
R61
100K
U19
5V_SW
TP29
C32
10nF
R63
100K
DMP2022LSS-13
U20B
SI1024X-T1-GE3
5V_SW_EN
R65
0R
GND
+12V_IN
+12V_START
CAN1_GND_ISO
GND
12V_GND_IN
+12V_IN
+12V_START
CAN1_GND_ISO
GND
12V_GND_IN
3.3V_SW_EN
5V_SW_EN
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO11
GPIO12
GPIO13
3.3V_SW_EN
5V_SW_EN
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO11
GPIO12
GPIO13
RH_UP
RH_DOWN
RH_LEFT
CHARGE_ACTIVE
IGNITION_ACTIVE_1
IGNITION_ACTIVE
RH_RIGHT
RH_CENTER
DASHBOARD_POWER_SENSE
RH_UP
RH_DOWN
RH_LEFT
CHARGE_ACTIVE
IGNITION_ACTIVE_1
IGNITION_ACTIVE
RH_RIGHT
RH_CENTER
DASHBOARD_POWER_SENSE

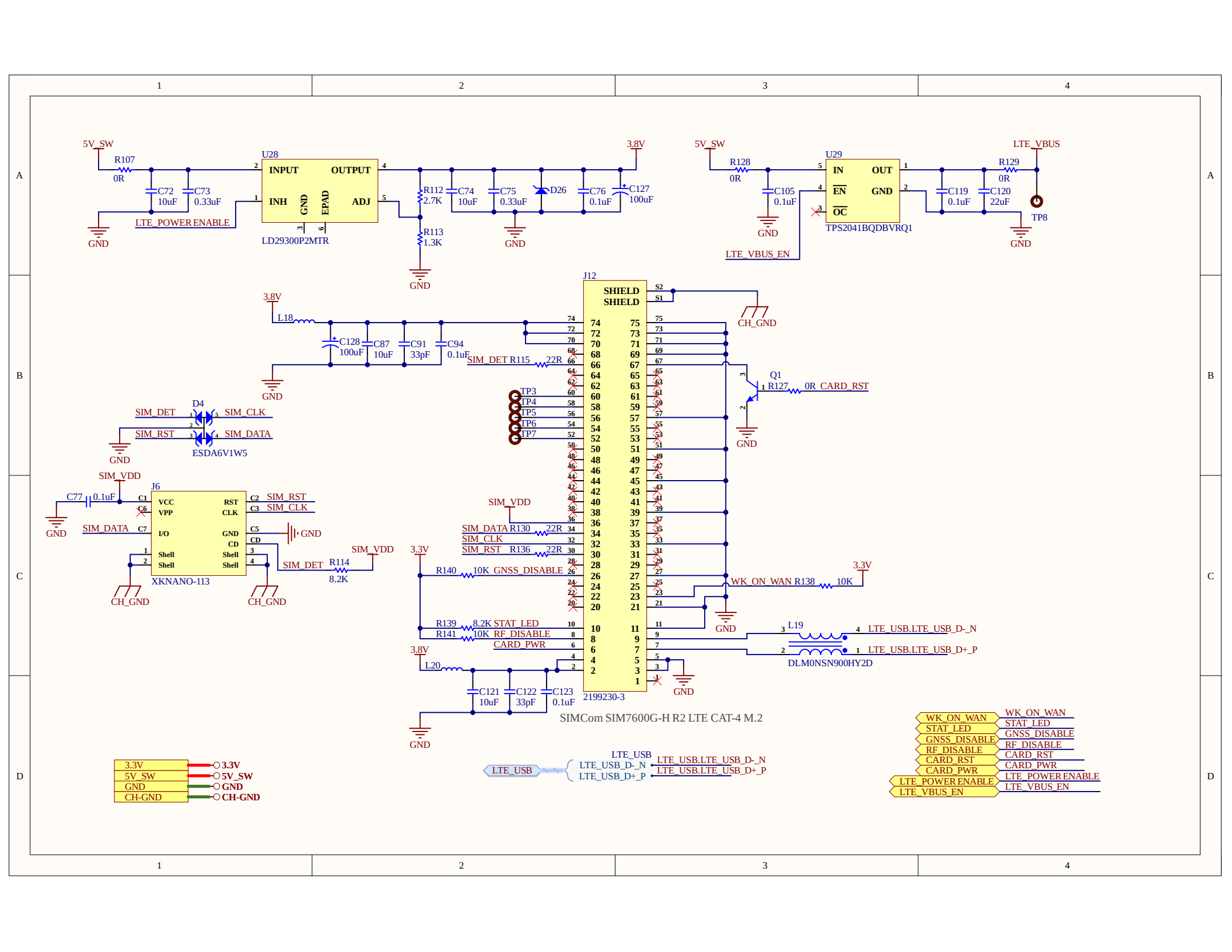

U21
S1
S2
S3
G
D
SQJ422EP-T1_GE3
U22
LM74610QDGKRQ1
Gate Pull Down
Gate Drive
Anode
Cathode
NC
NC
VcapL
VcapH
TP27
TP26
TP25
+12V_IN
F1
0458005.DR
D28
D29
C92
4.7uF
L28
12V_GND_IN
GND
C93
2.2uF
C88
4.7uF
C89
4.7uF
C90
4.7uF
L27
+12V_START
R37
2.7K
LED3
Emerald

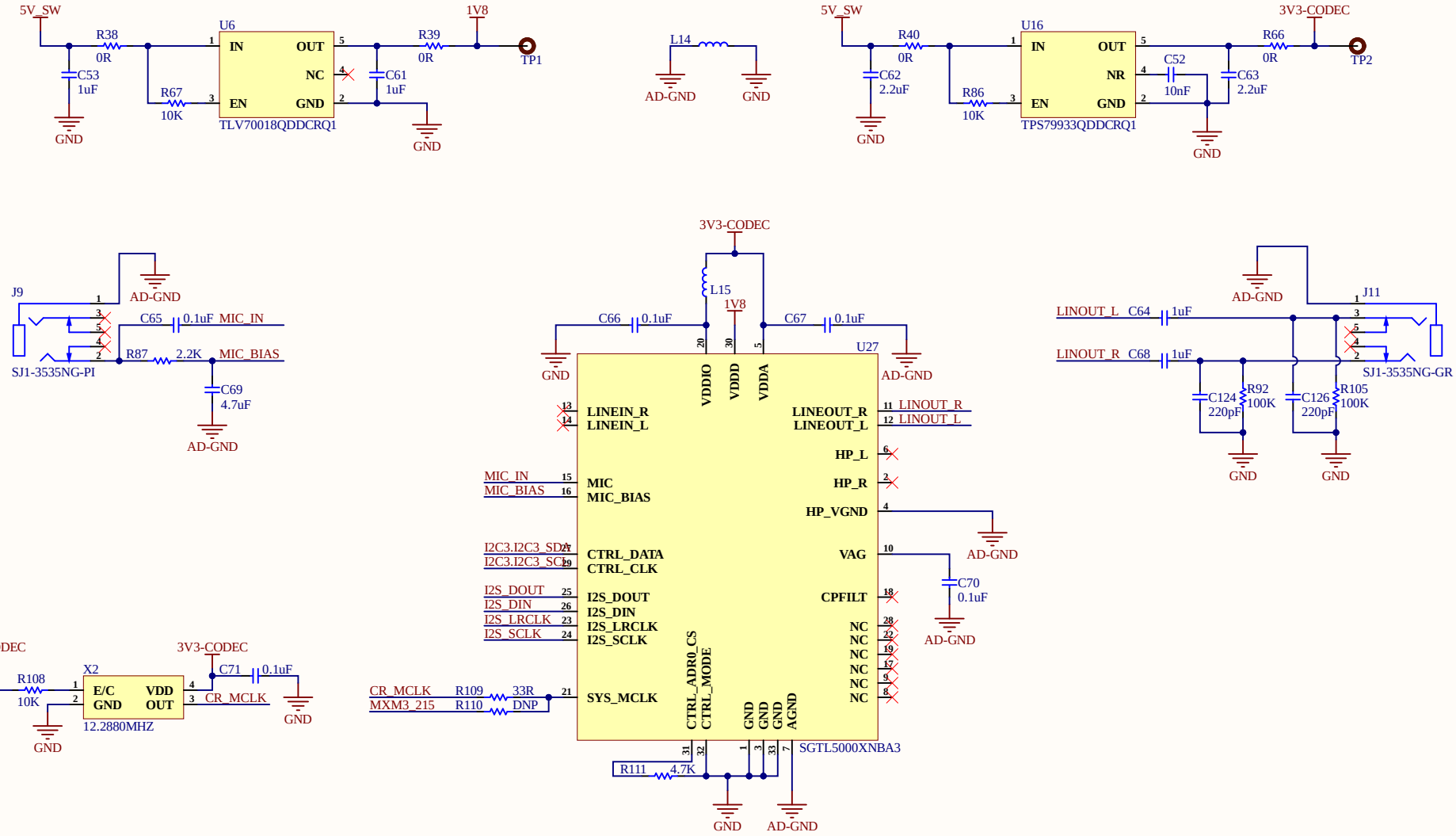

+12V_IN
+12V_START
GND
12V_GND_IN
+12V_IN
+12V_START
GND
12V_GND_IN


+12V_START
R137
0R
C144
0.1uF
C95
10uF
C96
0.1uF
GND
3.3V_BIAS
U24
SUPSW
SUP
EN
SYNC
BIAS
GND
RESET
FOSC
OUT
BST
LX
PGND
PGND
PGND
PGND
FB
COMP
MAX20006AFOB/VY+
R68
0R
C97
0.1uF
L16
1uH
3.3V
TP23
C98
47uF
C99
47uF
C100
0.1uF
R70
0R
R69
DNP
C101
2.2uF
R72
100K
TP30
R75
12K
R71
DNP
R35
330
LED1
RED
R73
DNP
C102
DNP
C103
3.9uF
R74
39K
C104
1000pF
+12V_START
R146
0R
C143
0.1uF
C106
10uF
C107
0.1uF
GND
5V_BIAS
U25
SUPSW
SUP
EN
SYNC
BIAS
GND
RESET
FOSC
OUT
BST
LX
PGND
PGND
PGND
PGND
FB
COMP
MAX20008AFOA/VY+
R76
0R
C108
0.1uF
L17
1uH
5V
TP24
C109
47uF
C110
47uF
C145
47uF
C146
47uF
C111
0.1uF
R78
0R
R77
DNP
C112
2.2uF
R80
100K
TP31
R82
12K
R79
DNP
R36
1K
LED2
RED
R81
DNP
C113
DNP
C114
3.9uF
R83
39K
C115
1000pF
3.3V
5V
+12V_START
GND


C116
1000pF
C117
1000pF
C118
1000pF
CH_GND
GND


GND
CH-GND
GND
CH-GND


5V_SW
R107
0R
C72
10uF
C73
0.33uF
LTE_POWER ENABLE
GND
U28
INPUT
OUTPUT
INH
GND
EPAD
ADJ
LD29300P2MTR
R112
2.7K
R113
1.3K
C74
10uF
C75
0.33uF
D26
C76
0.1uF
C127
100uF
3.8V
5V_SW
R128
0R
C105
0.1uF
LTE_VBUS_EN
U29
IN
OUT
EN
GND
OC
TPS2041BQDBVRQ1
C119
0.1uF
C120
22uF
R129
0R
LTE_VBUS
TP8
J12
SHIELD
SHIELD
CH_GND
3.8V
L18
C128
100uF
C87
10uF
C91
33pF
C94
0.1uF
SIM_DET R115
22R
TP3
TP4
TP5
TP6
TP7
Q1
R127
0R_CARD_RST
D4
SIM_DET
SIM_CLK
SIM_RST
SIM_DATA
ESDA6V1W5
SIM_VDD
J6
C77
0.1uF
VCC
VPP
I/O
RST
CLK
GND
CD
Shell
XKNANO-113
SIM_RST
SIM_CLK
SIM_DATA R130
22R
SIM_RST R136
22R
R140
10K GNSS_DISABLE
WK_ON_WAN R138
10K
R139
8.2K STAT_LED
R141
10K RF_DISABLE
CARD_PWR
L19
DLM0NSN900HY2D
LTE_USB.LTE_USB_D-_N
LTE_USB.LTE_USB_D+_P
L20
C121
10uF
C122
33pF
C123
0.1uF
R114
8.2K
2199230-3
SIMCom SIM7600G-H R2 LTE CAT-4 M.2
3.3V
5V_SW
GND
CH-GND
LTE_USB
LTE_USB_D-_N
LTE_USB_D+_P
WK_ON_WAN
STAT_LED
GNSS_DISABLE
RF_DISABLE
CARD_RST
CARD_PWR
LTE_POWER ENABLE
LTE_VBUS_EN


5V_SW
R38
0R
U6
IN
OUT
NC
EN
GND
TLV70018QDDCRQ1
C53
1uF
R67
10K
C61
1uF
R39
0R
1V8
TP1
GND
L14
AD-GND
5V_SW
R40
0R
U16
IN
OUT
NR
EN
GND
TPS79933QDDCRQ1
C62
2.2uF
R86
10K
C52
10nF
C63
2.2uF
R66
0R
3V3-CODEC
TP2
J9
SJ1-3535NG-PI
AD-GND
C65
0.1uF
MIC_IN
R87
2.2K
MIC_BIAS
C69
4.7uF
3V3-CODEC
L15
1V8
C66
0.1uF
C67
0.1uF
U27
VDDIO
VDDD
VDDA
LINEIN_R
LINEIN_L
LINEOUT_R
LINEOUT_L
LINOUT_R
LINOUT_L
HP_L
HP_R
HP_VGND
MIC
MIC_BIAS
CTRL_DATA
CTRL_CLK
VAG
C70
0.1uF
I2S_DOUT
I2S_DIN
I2S_LRCLK
I2S_SCLK
CPFILT
NC
SYS_MCLK
CTRL_ADR0_CS
CTRL_MODE
GND
AGND
SGTL5000XNBA3
R111
4.7K
J11
SJ1-3535NG-GR
LINOUT_L
LINOUT_R
C64
1uF
C68
1uF
C124
220pF
R92
100K
C126
220pF
R105
100K
X2
E/C
GND
VDD
OUT
12.2880MHZ
R108
10K
C71
0.1uF
CR_MCLK
MXM3_215
R109
33R
R110
DNP
I2C3.I2C3_SDA
I2C3.I2C3_SCL
5V_SW
GND
I2S_PORT
I2S_DOUT
I2S_DIN
I2S_LRCLK
I2S_SCLK
I2C3
I2C3_SCL
I2C3_SDA